

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“AI&ML Intern”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING

In

Department of Artificial Intelligence and Machine Learning

Submitted by:

Rahul Ramesh Yaragatti- 1DS20AI045

Conducted at



**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



CERTIFICATE

This is to certify that the Internship titled “**AI&ML Intern**” carried out by **Mr. RAHUL RAMESH YARAGATTI [1DS20AI045]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8ICINT**)

Signature of Reporting Head
Mohammed Mubin Mallick
CEO & Founder
Kiran Smart

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HOD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **RAHUL RAMESH YARAGATTI**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **KIRAN SMART**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023-2024.

Date: 24-04-2024

Place: Bangalore

USN: 1DS20AI045

NAME: RAHUL RAMESH
YARAGATTI

OFFER LETTER



KIRAN SMART

General Trading Co. WLL



Date : 15th Jan 2024

Dear Rahul Ramesh Yaragatti,

Subject: Internship Offer - Artificial Intelligence and Machine Learning (AI&ML) Intern

We are pleased to extend an offer for an internship position with **Kiran Smart** in the field of Artificial Intelligence and Machine Learning (AI&ML). We believe that your skills and enthusiasm make you an ideal candidate for this opportunity.

Internship Details: Student of Dayananda Sagar College of Engineering

Position: AI&ML Intern

Duration: 3 months

Start Date: February 1, 2024

Internship Overview:

During this internship, you will have the opportunity to work on cutting-edge projects in the field of Artificial Intelligence and Machine Learning. Under the mentorship of our experienced professionals, you will gain hands-on experience, contribute to real-world projects, and deepen your understanding of AI&ML applications in a business setting.

Key Responsibilities:

Collaborate with the AI&ML team on ongoing projects.

Contribute to the development and implementation of machine learning models.

Assist in data collection, preprocessing, and analysis.

Stay updated on the latest advancements in AI&ML technologies.

Participate in team meetings and contribute innovative ideas to solve complex problems.

We look forward to welcoming you to our team on February 1, 2024. To confirm your acceptance of this internship offer, please sign and return the attached copy of this letter by 25th Jan 2024.

We are excited about the contributions you will make to our team and the learning experiences that lie ahead. If you have any questions or require further information, please feel free to contact Ahmed Al Mutairi at info@kiransmart.com

Congratulations once again, and we anticipate a productive and enriching internship experience with you.

Sincerely

Mohammed Mubin Mallick (CEO & Founder)



Phone :
+965 97481976

Website :
www.kiransmart.com

Email :
info@kiransmart.com

Office # 1 - 8th Floor, Amaar Plaza Complex, Salmiya, Salem
Mubarak Street, Block 8, Building 71 - Kuwait

COMPLETION CERTIFICATE



KIRAN SMART

General Trading Co. WLL



Internship Completion Letter

Date: 3rd May 2024

This is to certify that **Rahul Ramesh Yaragatti** has successfully completed his **Internship** at Kiran Smart from 1st February 2024 to 30th April 2024.

During this period, he was working as an AI & ML Intern.

He has successfully completed all the tasks that were assigned to him. We wish him all the very best for his career ahead.

Regards,

Mohammed Mubin Mallick
Founder and CEO
Kiran Smart



Phone :
+965 97481976



Website :
www.kiransmart.com



Email :
info@kiransmart.com

Office # 1 - 8th Floor, Amaar Plaza Complex, Salmiya, Salem
Mubarak Street, Block 8, Building 71 - Kuwait

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Mohammed Mubin Mallick**, Kiran Smart, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

RAHUL RAMESH YARAGATTI
[1DS20AI045]

TABLE OF CONTENTS

Sl no	Description	Page no
1	INTRODUCTION	1
2	DESCRIPTION OF ORGANIZATION	2
3	SCOPE OF WORK	3
4	LEARNING OBJECTIVES AS GIVEN BY COMPANY	4
5	CHALLENGES AND SOLUTIONS	5
6	ACHIEVEMENTS AND CONTRIBUTION	6-7
7	REFLECTION AND ANALYSIS	8
8	RECOMMENDATIONS	9
	CONCLUSION	10
	REFERENCES	11

CHAPTER 1

INTRODUCTION

Kiran Smart is a leading robotics company with a multinational presence, including in the UAE (Dubai), Kuwait, and India (Bangalore). They specialize in humanoid robot solutions and are recognized globally for their innovative work.

Here are some key details about Kiran Smart:

Innovation: Kiran Smart is not limited by its solutions and services. It believes in innovation and provides smart solutions related to robotics, digital transformation, artificial intelligence (A.I.), Cyber-Security, High-Speed Internet Access (HSIA) Surfsonix, Meta-Verse, and Information Technological solutions.

Global Presence: Kiran Smart is based in the UAE (Dubai) and has a branch office in Kuwait and a registered office in India (Bangalore).

Services: Their services cater to a wide range of industries, including hotels, restaurants, hospitals, clinics, schools, banks, telecommunications, airports, shopping malls, hypermarkets, retail chain showrooms, automobiles, corporate offices, oil & gas, museums, and many more.

Products: They offer a variety of products, including interactive robots and guest pager solutions.

Mission: Their mission is to provide innovative, reliable, and high-quality services to clients throughout GCC and globally.

Vision: Their vision is to achieve global brand recognition as a finest solutions and services provider where customers find the optimal business solutions to their needs.

They were the first robotics company in the world to serve in an Iftar Party during Ramadan 2023. They are pioneers in robotics and humanoid solutions in Dubai, UAE, and Kuwait. They strive to offer state-of-the-art technology solutions that meet your budget requirements, ensuring you have dependable systems in the field of robotics, at competitive prices.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Kiran Smart is a leading multinational robotics company with a strong presence in the UAE (Dubai), Kuwait, and India (Bangalore). They specialize in providing innovative and smart solutions related to robotics, digital transformation, artificial intelligence (A.I.), Cyber-Security, High-Speed Internet Access (HSIA) Surfsonix, Meta-Verse, and Information Technological solutions. Their services cater to a wide range of industries, including hotels, restaurants, hospitals, clinics, schools, banks, telecommunications, airports, shopping malls, hypermarkets, retail chain showrooms, automobiles, corporate offices, oil & gas, museums, and many more. They offer a variety of products, including interactive robots and guest pager solutions.

Kiran Smart's mission is to provide innovative, reliable, and high-quality services to clients throughout GCC and globally. Their vision is to achieve global brand recognition as a finest solutions and services provider where customers find the optimal business solutions to their needs.

Some of the organization's notable work include as follows:

Project Execution: Kiran Smart has a professional team that gets in touch with the customer to understand the details of the project requirement. Once the project is defined, their solution architect expert prepares the most innovative solution keeping in mind the cost and other factors. Their team communicates with the customer on every stage of development of the project to show the progress and deliver on time.

Product Range: They offer a wide range of products including interactive robots, guest pager solutions, and various digital screens and tables.

Why Choose Kiran Smart: They have vast experience and techno know-how of robotic solutions. They deal with multiple robotic manufacturers and companies around the world to deliver the right solution to their customers.

Industries Served: As industries increasingly embrace robot solutions and services, Kiran Smart leads the way with its innovative humanoid robot solutions. They cater to a wide range of industries, including hotels, restaurants, hospitals, clinics, schools, banks, telecommunications, airports, shopping malls, hypermarkets, retail chain showrooms, automobiles, corporate offices, oil & gas, museums, and many more industries currently in development.

CHAPTER 3

SCOPE OF WORK

During the internship period at Kiran Smart company, the scope of work encompassed two main phases: AI&ML project development and web development.

1. **Web Development Phase:** Initially, the scope involved tasks related to web development utilizing technologies such as HTML, CSS, JavaScript, Bootstrap, and JQuery.

The objectives during this phase included:

- Understanding the basics of front-end web development.
- Practicing the implementation of responsive and interactive web interfaces.
- Gaining familiarity with popular web development frameworks and libraries.
- Applying acquired skills to contribute to ongoing web projects within the company.

2. **AI & ML Project Development Phase:** Following the successful completion of the web development tasks, the scope transitioned to a project task focused on the development of a Healthcare Chatbot named Sultan. This project required proficiency in Python, Deep Learning fundamentals (specifically TensorFlow & Keras), NLP (Natural Language Processing), Flask, HTML, CSS, JavaScript, and SQLite.

The objectives and scope of work for this phase were as follows:

- Developing a Healthcare Chatbot named Sultan using deep learning techniques.
- Integrating the Chatbot into the company's robotics system for seamless hospital services.
- Leveraging Python for backend development, including the implementation of deep learning algorithms.
- Utilizing TensorFlow and Keras for building and training neural network models.
- Implementing NLP techniques to enable the Chatbot to understand and respond to natural language queries effectively.
- Integrating the Chatbot with a Flask-based web interface for user interaction.
- Incorporating HTML, CSS, and JavaScript to enhance the user experience of the Chatbot interface.
- Employing SQLite for managing the Chatbot's database and storing relevant information.
- Ensuring compatibility with the SpeechRecognition module to enable voice-based interactions with the Chatbot.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

During the internship at Kiran Smart company, the company outlined specific learning objectives aimed at fostering the intern's growth and skill development in relevant domains. These learning objectives were tailored to align with the intern's role and the projects assigned.

The identified learning objectives were as follows:

- 1. Web Development Proficiency:** Gain a solid understanding of front-end web development technologies, including HTML, CSS, JavaScript, Bootstrap, and JQuery. Learn to create responsive and visually appealing web interfaces. Acquire proficiency in utilizing web development frameworks and libraries for efficient development practices. Develop the ability to contribute effectively to ongoing web projects within the company.
- 2. Deep Learning and AI Application:** Develop proficiency in Python programming language, particularly for AI and ML applications. Acquire a strong understanding of deep learning fundamentals, with a focus on TensorFlow and Keras frameworks. Learn to apply deep learning techniques to real-world projects, particularly in the development of AI-driven solutions. Gain practical experience in implementing NLP (Natural Language Processing) algorithms for text-based interactions. Understand the integration of AI models into web applications using frameworks like Flask. Familiarize oneself with database management systems, particularly SQLite, for data storage and retrieval within AI applications.
- 3. Project Management and Integration:** Develop project management skills by working on structured project tasks from initiation to completion. Learn to effectively integrate developed solutions into existing company systems, such as robotics systems, for enhanced functionality and service delivery. Gain experience in collaborating with cross-functional teams, including developers, designers, and project managers, to achieve project objectives within specified timelines. Understand the importance of documentation and version control in maintaining project integrity and facilitating knowledge transfer within the organization.

CHAPTER 5

CHALLENGES AND SOLUTIONS

Throughout the internship at Kiran Smart company, several challenges were encountered during the execution of tasks and projects. However, these challenges provided valuable learning opportunities and prompted the development of effective solutions.

1. Technical Complexity:

Challenge: Developing a Healthcare Chatbot utilizing deep learning techniques posed significant technical complexity, particularly in integrating various technologies and frameworks.

Solution: To address this challenge, a systematic approach was adopted, starting with thorough research and understanding of each technology involved. Breakdown of tasks into manageable components and continuous collaboration with mentors and team members facilitated step-by-step progress. Additionally, seeking guidance from online resources, documentation, and relevant tutorials helped in navigating through technical intricacies.

2. Resource Limitations:

Challenge: Limited access to specialized hardware resources for training deep learning models, such as GPUs, posed a challenge in achieving optimal performance within project constraints.

Solution: To mitigate this challenge, alternative solutions were explored, including cloud-based services offering GPU instances for model training. Leveraging cloud platforms such as Google Cloud Platform or AWS (Amazon Web Services) enabled access to scalable computing resources, thereby overcoming limitations associated with hardware constraints. Additionally, optimizing model architectures and hyperparameters helped in maximizing the efficiency while minimizing the resource usage.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

Task 1:

HTML Web Page Development - Developed an online form incorporating fields for personal information, gender, DOB, contact details, vehicle ownership, color selection, document upload, and comments. Implemented validation, clean layout, and adherence to company branding.

The screenshot displays a web page with a logo and a search bar at the top. Below the logo is a navigation menu with links: Form, Table, List, and Heading. The main content area is divided into two sections. The left section, titled 'Form', contains various input fields: First Name (placeholder: 'Please enter your first name'), Last Name (placeholder: 'Please enter your last name'), Gender (radio buttons for Male and Female), DOB (placeholder: 'dd-mm-yyyy' with a calendar icon), Email, Telephone, I have (radio buttons for Bike and Car), Select your favorite color (a color picker), Upload your documents (Choose File button, No file chosen text), and Comments (text area). At the bottom of the form are Submit and Reset buttons. The right section contains a table with three columns: First Name, Last Name, and Email. The table has two rows of data: Jill Smith with email jillsmith@gmail.com, and Eve Jackson with email evejackson@gmail.com. Below the table is a section titled 'Heading Tags' with examples of heading levels from 1 to 6. To the right of the heading tags is a section titled 'List' with two examples: 'An Unordered List' (ul) containing Coffee, Tea, and Milk, and 'An Ordered List' (ol) containing 1. Coffee, 2. Tea, and 3. Milk.

First Name	Last Name	Email
Jill	Smith	jillsmith@gmail.com
Eve	Jackson	evejackson@gmail.com

Figure 6.1 - HTMLTask

Task 2:

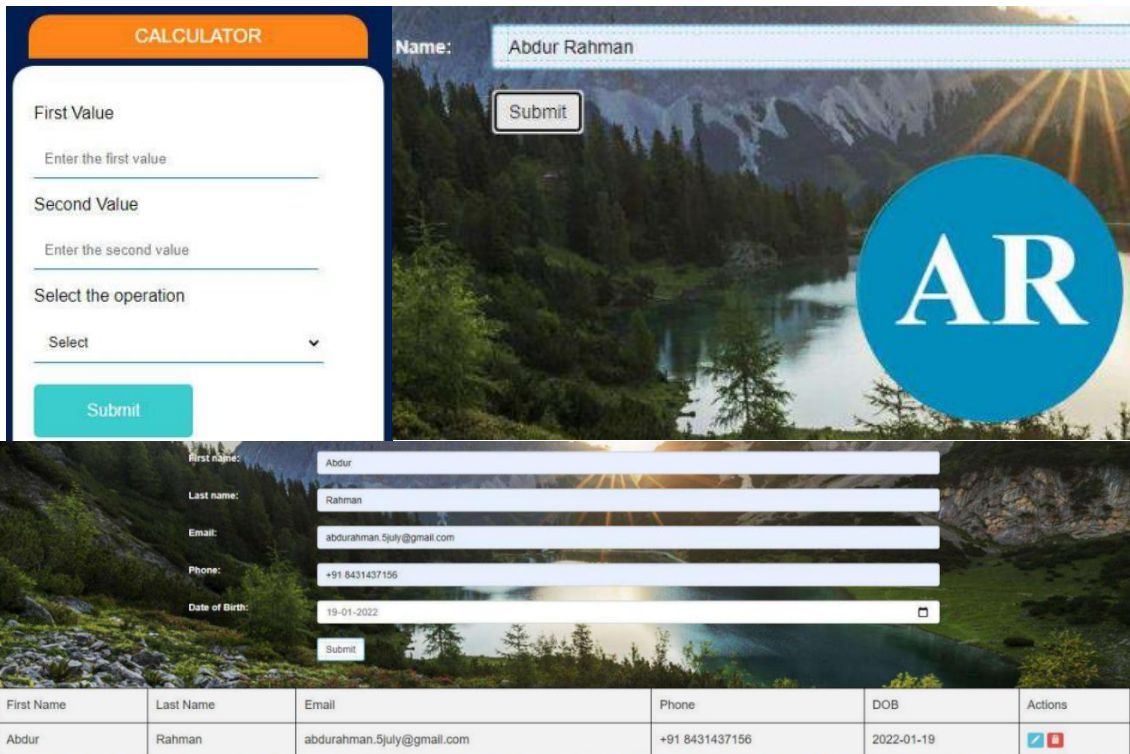
CSS and Bootstrap Implementation- Built various Navbars, Forms (Default, Inline, and Horizontal), Tables with searchable fields, different button categories (sizes, badge, and dropdown), and an alert message.

The screenshot shows a web page with a teal navbar at the top containing links: Navbar, Forms, Table, Button, and Modal. Below the navbar is a horizontal list of links: WebSiteName, Home, Page 1, Page 2, Sign Up, and Login. The main content area is divided into two sections. The left section shows a collection of buttons with different styles: Basic, Default, Primary, Success, Info, Warning, Danger, Link, Active, and Disabled. The right section shows a form with fields for Email address and Password, a Remember me checkbox, and a Submit button. The form is styled with a light blue background and rounded corners.

Figure 6.2 – CSS and Bootstrap Task

Task 3:

JavaScript and jQuery Functionality - Utilized JavaScript and jQuery for DOM manipulation and Bootstrap/CSS for beautification. Implemented functionality including arithmetic calculations, initials generation, form-to-table conversion, table-to-form population, and row deletion.



The screenshot displays a web application with two main sections. The top section, titled 'CALCULATOR', contains input fields for 'First Value' and 'Second Value', a dropdown for 'Select the operation', and a 'Submit' button. The bottom section features a form for generating initials, with fields for 'First name' (Abdur), 'Last name' (Rahman), 'Email' (abdurahman.5july@gmail.com), 'Phone' (+91 8431437156), and 'Date of Birth' (19-01-2022). A large blue circle with the initials 'AR' is prominently displayed on the right side of the form. Below the form is a table with columns for First Name, Last Name, Email, Phone, DOB, and Actions.

First Name	Last Name	Email	Phone	DOB	Actions
Abdur	Rahman	abdurahman.5july@gmail.com	+91 8431437156	2022-01-19	 

Figure 6.3 - Javascript and JQuery Task

Task 4

Deep Learning Chatbot Development - Developed a Healthcare Chatbot named Sultan using deep learning techniques (TensorFlow & Keras) for integration into the company's robotics system. Implemented NLP, Flask, HTML, CSS, JavaScript, and SQLite. Designed a Seq2Seq model for disease prediction from symptoms.



The screenshot shows a chatbot interface with a dark background and white text. The chatbot's name 'chatbot' is visible in the top left corner. The conversation history includes a timestamp '2022-07-21 13:20:17.294369', a system message 'I tensorflow/core/platform/cpu_fe', and a user input 'To enable them in other operations, rebuild TensorFlow with t'. The chatbot responds with 'Bot is Running', 'Say Your Symptoms', and '1/1 [=====] - 0s 188ms/step'. The user then says 'You said : I am suffering from Rapid Heartbeat and sweating', and the chatbot responds with 'Result : You have Anxiety or Panic Disorder' and 'Say Your Symptoms'.

Figure 6.4 – Deep Learning Chatbot Task

CHAPTER 7

REFLECTION AND ANALYSIS

- ❖ **Personal Growth:** Reflecting on how the internship has contributed to personal and professional growth, including improvements in technical skills, problem-solving abilities, and confidence in tackling challenges.
- ❖ **Project Experiences:** Analyzing key project experiences, highlighting successes, challenges overcome, and lessons learned throughout the development process.
- ❖ **Mentorship Impact:** Examining the impact of mentorship on learning and development, acknowledging the guidance received and the value of constructive feedback in shaping skills and perspectives.
- ❖ **Collaborative Learning:** Reflecting on the collaborative learning environment within the team, discussing the benefits of teamwork, communication, and knowledge sharing in achieving project goals.
- ❖ **Future Goals:** Outlining future goals and aspirations in the field of deep learning (AI&ML) and web development, considering areas for continued learning, skill enhancement, and professional growth beyond the internship experience.

CHAPTER 8

RECOMMENDATIONS

1. Structured Learning Path:

Design a structured learning path that covers essential concepts, technologies, and best practices in deep learning (AI&ML) and web development. Provide a curriculum or syllabus outlining the topics to be covered, along with recommended resources such as online courses, tutorials, and documentation.

2. Hands-on Projects:

Offer hands-on projects that allow interns to apply their knowledge and skills in real-world scenarios. Assign projects of varying complexity to cater to different skill levels and interests, and provide guidance and support as needed.

3. Mentorship and Guidance:

Pair each intern with a mentor who can provide guidance, support, and feedback throughout the internship. Encourage mentors to share their expertise, provide technical assistance, and offer career advice to help interns succeed.

4. Collaboration Opportunities:

Foster a collaborative environment where interns can work together on projects, share ideas, and learn from each other. Encourage teamwork, communication, and knowledge sharing through pair programming, code reviews, and collaborative problem-solving sessions.

5. Networking and Professional Development:

Facilitate networking opportunities for interns to connect with industry professionals, attend meetups or conferences, and build their professional network. Offer workshops, seminars, or guest lectures on topics related to career development, industry trends, and emerging technologies.

6. Recognition and Celebration:

Recognize and celebrate interns' achievements, milestones, and contributions throughout the internship. Showcase interns' work through presentations, demos, or portfolio showcases to highlight their skills and accomplishments.

CONCLUSION

In conclusion, the internship at Kiran Smart company has been a transformative experience, marked by significant personal and professional growth. Over the course of the internship, I have had the opportunity to work on diverse projects spanning web development, AI, and ML, each presenting unique challenges and learning opportunities. From building HTML web pages to developing a sophisticated Healthcare Chatbot named Sultan, I have honed my technical skills and gained hands-on experience in cutting-edge technologies. The challenges encountered along the way, including technical complexity, resource limitations, and time constraints, have been overcome through perseverance, collaboration, and innovative problem-solving. Moreover, the mentorship and support provided by colleagues and supervisors have been invaluable in navigating the internship journey. Looking ahead, I am excited to apply the lessons learned and skills acquired during the internship to future endeavors, as I continue to pursue excellence in the field of technology and innovation. I am immensely grateful for the opportunity to intern with Kiran Smart company and am eager to contribute meaningfully to the organization's success in the future.

REFERENCES

1. www.kiransmart.com
2. <https://theleadersglobe.com/magazine/kiran-smart-the-brand-pioneering-robotic-solutions-for-the-everyday/>
3. <https://businesschiefsinsight.com/kiran-smart-delivering-the-most-innovative-and-smart-solutions-for-all-industries/>
4. www.tutorialspoint.com
5. <https://www.geeksforgeeks.org/>